Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection

R26-IT-011

Project Proposal Report

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BSc (Hons) in Information Technology – Information Technology

Department of Information Technology

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March 2025

Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection - Age & Gender Detection using Face-API

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BSc (Hons) in Information Technology – Information Technology

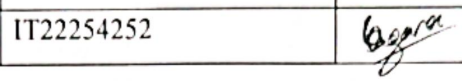
Department of Information Technology

Sri Lankan Institute of Information Technology

March 2025

Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	**Student ID**	**Signature**
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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for their undergraduate dissertations under my supervision.

Dr. Junius Anjana

Signature of the supervisor:  19th /Mar /2026 Date: ~~03/13/2026~~

Abstract

This research project focuses on developing an intelligent web-based facial analysis system capable of detecting human age and gender using modern computer vision techniques. The system utilizes deep learning models implemented through the JavaScript library Face-api.js, which runs directly in the web browser using TensorFlow.js. By capturing real-time video through a webcam, the application detects faces and estimates demographic attributes such as approximate age and gender. The primary goal of the project is to evaluate the feasibility of performing accurate facial attribute analysis using lightweight AI models without relying on external cloud services. This approach enhances user privacy while maintaining real-time performance and accessibility across common web browsers. The proposed system aims to demonstrate how browser-based artificial intelligence can be applied to practical applications such as smart interfaces, demographic analytics, and human-computer interaction systems. Through performance evaluation and testing under different environmental conditions, the project will analyze the accuracy, efficiency, and usability of the developed solution.

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List of abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CVAI	Cranial Vault Asymmetry Index
CI	Cephalic Index
mHealth	Mobile Health
UI	User Interface
UX	User Experience
SUS	System Usability Scale
IDE	Integrated Development Environment
API	Application Programming Interface
SQLite	Structured Query Language Lite
SDK	Software Development Kit
BaaS	Backend as a Service
FOSS	Free and Open-Source Software
LKR	Sri Lankan Rupees
NLP	Natural Language Processing
RGB	Red, Green, Blue (color model used in image processing)

ROI	Region of Interest (used in image analysis)
FPS	Frames Per Second
UI/UX	User Interface / User Experience
ML	Machine Learning
API	Application Programming Interface

Introduction

With the rapid advancement of Artificial Intelligence and Computer Vision technologies, facial analysis systems have become an important area of research and development. Modern AI models are capable of analyzing facial features to extract useful information such as age, gender, emotions, and identity. These technologies are widely used in areas such as human–computer interaction, smart security systems, and demographic analytics.

This project focuses on developing a web-based facial analysis system that can detect and estimate a person's age and gender in real time using computer vision techniques. The system uses Face-api.js, a JavaScript library built on top of TensorFlow.js, which allows deep learning models to run directly inside a web browser. By using a webcam as input, the application detects faces and predicts demographic attributes without sending data to external servers.

The goal of this project is to demonstrate how lightweight browser-based AI models can perform real-time facial analysis efficiently while maintaining user privacy and accessibility. The developed system will evaluate the performance, accuracy, and usability of age and gender detection under different environmental conditions such as lighting, camera quality, and facial orientation.

Background & Literature survey

Face analysis has become an important research area in the field of Computer Vision and Artificial Intelligence. Modern AI systems are able to analyze facial images to estimate attributes such as age, gender, emotions, and identity. These technologies are widely used in smart surveillance, marketing analytics, and human–computer interaction systems.

One of the most widely used tools for browser-based facial analysis is Face-api.js. It is a JavaScript library built on top of TensorFlow.js that allows developers to run deep learning models directly in web browsers. This makes it possible to perform face detection and demographic estimation without sending images to a remote server, which improves privacy and reduces processing delays.

Several studies have explored deep learning models for predicting age and gender from facial images. For example, convolutional neural networks (CNNs) have shown strong performance in recognizing facial features and learning patterns associated with age groups and gender characteristics. CNN models trained on large datasets such as IMDB-WIKI Dataset and Adience Dataset have demonstrated high accuracy in predicting age ranges and gender categories.

Another important approach involves lightweight deep learning architectures designed for real-time applications. These models can run efficiently on web browsers and mobile devices without requiring powerful hardware. Libraries such as Face-api.js include pre-trained CNN models that estimate age as a numerical value and gender as a probability distribution (male or female).

Research has shown that browser-based AI solutions are increasingly popular because they provide fast results while maintaining user privacy. Since image data does not need to be transmitted to external servers, the risk of sensitive data exposure is minimized. These advantages make browser-based facial analysis systems suitable for educational research projects and lightweight AI applications.

Overall, the literature shows that AI-based facial analysis using CNN models and browser-based frameworks can provide accurate and efficient age and gender estimation.

However, further research is needed to evaluate how these technologies perform in real-time web applications and under different environmental conditions such as lighting changes, camera quality, and facial angles.

Research Gap

Although facial analysis technologies have significantly improved with the development of modern Artificial Intelligence techniques, several limitations still exist in practical implementations of age and gender detection systems. Many current solutions are developed mainly for controlled research environments or commercial systems with powerful computing resources. As a result, these systems may not perform consistently when applied in lightweight web-based applications or real-time environments using standard consumer devices such as webcams or mobile cameras.

One major gap in existing research is the heavy reliance on cloud-based processing. Many facial analysis applications send captured images to remote servers where deep learning models process the data and return predictions. While this method can provide high accuracy due to the use of powerful servers and large models, it raises concerns related to privacy, data security, and internet dependency. Users may feel uncomfortable sharing facial data with external servers, especially in applications involving personal identification or demographic analysis. Additionally, real-time systems that rely on network communication may experience delays or interruptions when internet connectivity is weak or unavailable.

Another limitation found in existing systems is the lack of lightweight implementations that can operate efficiently within web browsers. Most deep learning models used for facial analysis are large and computationally expensive, making them difficult to deploy on client-side environments. As a result, many studies focus primarily on model accuracy without considering the practical challenges of integrating these models into accessible web applications. This creates a gap between theoretical research and real-world implementation.

Furthermore, many datasets used for training age and gender prediction models are collected under controlled conditions with high-quality images and balanced demographic distributions. In real-world scenarios, facial images captured through webcams may suffer from variations in lighting, camera resolution, facial orientation, and background noise. These environmental factors can significantly reduce prediction accuracy. Therefore, additional research is needed to evaluate how facial analysis models perform under everyday conditions where image quality is less predictable.

Another gap relates to the limited evaluation of real-time user interaction within facial analysis systems. Many existing studies focus mainly on algorithm performance while giving less attention to usability, system responsiveness, and user experience. For web-based applications, factors such as processing speed, interface clarity, and system stability are critical to ensure smooth interaction between the user and the AI system.

To address these gaps, this research proposes a browser-based facial analysis system that performs age and gender detection directly on the client side using lightweight deep learning models. By utilizing modern web technologies and efficient AI frameworks, the proposed system aims to evaluate whether real-time facial analysis can be performed accurately and efficiently without relying on external cloud services. This approach helps improve privacy protection, reduces processing delays, and demonstrates the potential of integrating artificial intelligence directly into web applications.

Table 1 Comparative Table

Area	**Existing Studies / Tools**	**Strengths**	**Gaps**
Age & Gender Detection	Face-api.js	- Lightweight deep learning models - Runs directly in web browsers -	- Accuracy affected by lighting and camera quality - Limited evaluation in real-world environments

Area	Existing Studies / Tools	Strengths	Gaps
		Real-time face detection and attribute estimation	
Deep Learning Models	CNN-based models trained on IMDB-WIKI Dataset and Adience Dataset	- High accuracy in predicting age groups and gender - Large-scale datasets used for training	- Datasets contain controlled images - Limited diversity in real-world conditions
Cloud-Based Facial Analysis	Microsoft Azure Face API, Amazon Rekognition	- High processing power - High prediction accuracy	- Requires internet connection - Privacy concerns due to cloud processing
Real-Time Web Applications	Browser-based AI frameworks such as TensorFlow.js	- Allows AI models to run in web browsers - Reduces server dependency	- Performance varies across devices - Limited research on real-time browser implementations
Privacy & Data Processing	Client-side AI processing	- Improves user privacy - No need to upload facial images to servers	- Limited studies on accuracy and performance trade-offs

Research Problem

Facial analysis technologies have advanced significantly in recent years, but many real-time applications still face limitations related to performance, privacy, and accessibility. Most existing age and gender detection systems rely on cloud-based AI services or high-performance computing resources. These solutions require constant internet connectivity

that utilizes the capabilities of Face-api.js, a JavaScript library built on top of TensorFlow.js. By capturing video input from a user's webcam, the system will analyze facial features and provide predictions related to age and gender in real time.
The proposed system aims to demonstrate how lightweight deep learning models can operate directly within a web browser without requiring external cloud processing. This approach enhances user privacy while ensuring efficient processing and accessibility across different devices and browsers. Additionally, the project seeks to evaluate the accuracy, responsiveness, and usability of browser-based facial analysis systems when used under different real-world conditions such as lighting variations, camera resolution differences, and multiple face detection scenarios.

Specific Objectives

1. Implement Face Detection using Face-api.js

The first objective of the project is to implement a reliable face detection mechanism using the Face-api.js framework. This component will allow the system to identify and locate human faces within a real-time video stream captured from a webcam. The pre-trained face detection models provided by Face-api.js will be used to detect facial regions and generate bounding boxes around detected faces. This step is essential because accurate face detection forms the foundation for further facial analysis tasks such as age estimation and gender classification.

2. Develop an Age Prediction Module

Another important objective is to develop a module that can estimate the approximate age of detected individuals. The system will use deep learning models trained on large facial datasets to analyze facial features and patterns associated with different age

groups. Once a face is detected, the model will process the facial image and generate an estimated age value. This module will allow the system to provide real-time age predictions while maintaining efficient processing within the browser environment.

3. Develop a Gender Classification Module

The project will also include the development of a gender classification component that identifies the likely gender of detected individuals. This module will use trained neural network models capable of recognizing facial characteristics that are commonly associated with gender categories. The system will analyze the detected facial region and classify it into gender classes with associated probability values. This feature enhances the overall functionality of the facial analysis system and demonstrates the capability of AI-based demographic prediction.

4. Integrate the Model into a Web Interface

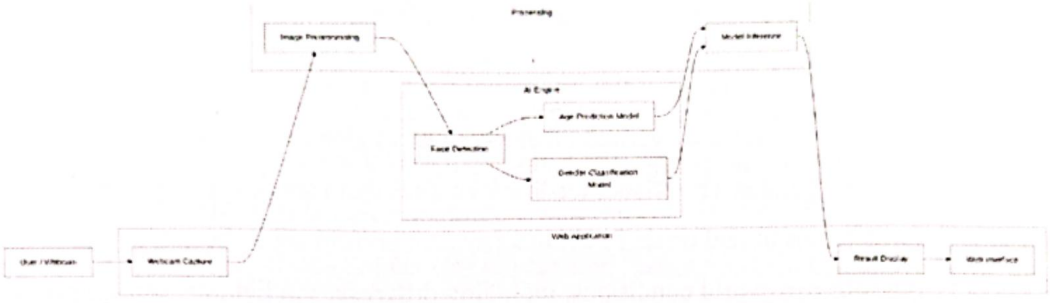
Another objective of the project is to integrate the facial analysis models into an interactive web-based interface. The application will provide a simple and user-friendly interface that allows users to access their webcam and view real-time analysis results. When faces are detected in the video stream, the system will display visual indicators such as bounding boxes and labels that show the predicted age and gender. This integration ensures that the system is not only technically functional but also accessible and easy for users to interact with.

5. Evaluate System Performance

The final objective of the project is to evaluate the performance of the developed system under different conditions. This evaluation will involve testing the system using various lighting environments, camera qualities, and facial orientations to observe how these

factors affect prediction accuracy and system responsiveness. Performance metrics such as processing speed, accuracy of predictions, and overall system stability will be analyzed to determine the effectiveness of the implemented solution.

Figure 1 System Diagram



Methodology

The development of the Age and Gender Detection system follows a structured methodology designed to integrate browser-based AI with real-time facial analysis. The process begins with an extensive literature review aimed at understanding the current state of facial analysis and demographic prediction. This review focuses on convolutional neural network (CNN)-based approaches for age and gender estimation, as well as browser-compatible AI frameworks such as Face-api.js and TensorFlow.js. Existing facial datasets, including IMDB-WIKI, FG-NET, and MORPH, are also analyzed to understand the data distribution, labeling, and model training techniques that have proven effective in previous studies. The insights gathered from this stage form the foundation for designing an efficient and accurate system architecture.

The system architecture is designed to operate entirely within a web browser, leveraging client-side processing to ensure fast, real-time predictions. The application captures live

video input through the browser's webcam APIs, which is then processed using pre-trained face detection models available in Face-api.js. These models identify facial regions in the video stream, creating bounding boxes around detected faces for further analysis. Once the faces are detected, the system applies age and gender estimation models, which have been trained on large-scale facial datasets and optimized for in-browser inference using TensorFlow.js. The models output predicted age values as well as gender classifications for each detected face.

The results from the prediction models are then rendered on the user interface, where bounding boxes are overlaid on detected faces along with labels indicating the estimated age and gender. This visual feedback ensures that users can easily interpret the predictions in real time. To evaluate system performance, the application is tested under various real-world conditions, including differences in lighting, camera resolution, facial orientation, and background complexity. Performance metrics such as prediction accuracy, processing speed, and robustness to challenging scenarios are recorded and analyzed.

Finally, the results are systematically analyzed to assess the effectiveness of browser-based AI for facial demographic analysis. Comparative analysis is conducted against reported benchmarks in the literature to understand the model's strengths and limitations. The methodology ensures a comprehensive approach, combining theoretical insights, practical system design, and empirical evaluation, to produce a reliable and responsive age and gender detection system suitable for deployment in browser-based applications.

Functional Requirements

Functional requirements describe what your system must do—the core features that enable midwives to use the app effectively in real-world conditions.

1. User Authentication and Access - This ensures that only authorized users (e.g., midwives) can access the app. It protects sensitive health data and allows for future role-based access, such as adding supervisors or researchers.
2. Image and Video Capture - Midwives must be able to capture or upload images/videos of infants directly through the app. This feature is central to enabling AI analysis and must include visual guidance (e.g., framing grids) to ensure diagnostic accuracy.
3. AI-Powered Analysis - The app must process visual input using the embedded AI model to detect cranial deformities and postural irregularities. It should output clinical metrics like CVAI, CI, and posture angles, along with easy-to-understand visual indicators (e.g., color-coded risk levels).
4. Bilingual Interface - All content navigation, instructions, and results must be available in Sinhala and English. This ensures inclusiveness for midwives across linguistic backgrounds and supports training in both urban and rural contexts.
5. Result Interpretation and Guidance - AI-generated results must be presented in a simplified format, using culturally adapted language and visual cues. The app should offer actionable recommendations to support decision-making.
6. Audio-Visual Storytelling Integration – To improve understanding and interactivity, the app must play short, pre-recorded Sinhala/English audio narrations and display simple looping motion graphics that explain the detected result and recommended action. This helps midwives, including those with lower literacy, interpret results quickly and confidently.
7. Usability Feedback Collection - The app should include a built-in feedback module, allowing midwives to rate usability (e.g., SUS scores) and provide comments. This feedback loop supports iterative design and ensures the app evolves with user needs.

8. Data Privacy and Security - All personal and health data must be encrypted during storage and transmission. Users should have control over what data is stored, shared, or deleted, in compliance with Sri Lankan data protection laws.

Non-Functional Requirements

Non-functional requirements define how the system behaves rather than what it does. For the Age & Gender Detection system, these requirements focus on performance, usability, security, and reliability to ensure that the web-based facial analysis application is effective and user-friendly in real-world conditions.

1. **Performance**
    - The system should process video frames from the webcam in real time, providing age and gender predictions with minimal delay (ideally under 2 seconds per frame).
    - The application should handle multiple faces simultaneously without noticeable degradation in performance.
    - Efficient use of browser resources should be ensured to prevent excessive CPU or memory usage.
2. **Usability**
    - The web interface should be intuitive and easy to navigate for users with varying levels of technical expertise.
    - Detected faces and corresponding predictions (age and gender) should be clearly displayed with visual indicators such as bounding boxes, labels, or color codes.

- Instructions or guidance (if needed) should be concise and visually clear to avoid confusion.

3. **Compatibility**
    - The system should be compatible with major modern browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge.
    - The system should operate on standard consumer devices such as laptops, desktops, and tablets without requiring dedicated hardware.
4. **Privacy and Security**
    - All facial image processing must occur locally within the user's browser; no images should be sent to external servers unless explicitly required.
    - Any temporary or stored data should be handled securely, with the ability to delete local caches or saved results at any time.
    - The system should comply with general data privacy guidelines to protect sensitive user information.
5. **Reliability and Robustness**
    - The system should maintain stable operation under varying lighting conditions, camera resolutions, and facial orientations.
    - Error handling should be implemented to manage cases where no face is detected, multiple faces overlap, or input is invalid.
    - The system should recover gracefully from interruptions, such as loss of webcam access or browser refresh.

6. **Scalability**
    - While the system primarily targets single or small group usage, the architecture should allow future expansion, such as adding additional demographic predictions or emotion detection.
    - Modular design should allow new AI models or features to be integrated without rewriting the core system.
7. **Maintainability**
    - The codebase should be structured with clear separation of modules (e.g., UI, AI engine, preprocessing) to allow easy updates and debugging.
    - Proper documentation should be provided for future developers to understand system components and extend functionality.
8. **Ethical and Safe Design**
    - Predictions should include disclaimers indicating that results are estimations and should not replace professional judgment.
    - The system should avoid biased predictions and strive for fairness across different age ranges, genders, and ethnic backgrounds where possible.

System Requirements

System requirements define the technical specifications needed to develop, deploy, and operate a software solution effectively. They ensure that the application runs smoothly, remains secure, and performs reliably across different environments. These requirements are typically divided into two categories:

1. Hardware Requirements: The physical devices and infrastructure needed (e.g., smartphones, computers).

2. Software Requirements: The platforms, frameworks, and tools used to build and run the application.

Hardware Requirements

1. **User Devices**

- Desktop or laptop with webcam access.
- Minimum specifications: 2 GB RAM, 1.5 GHz CPU, standard integrated GPU.
- Higher-end devices recommended for better real-time performance with multiple faces.

2. **Development Machine**

- Operating System: Windows, macOS, or Linux.
- Minimum 8 GB RAM, 256 GB SSD.
- Optional: Dedicated GPU (NVIDIA or AMD) for faster model training and testing.

3. **Network**

- Internet access is optional for local execution but required for initial dependency installation or cloud-based model downloads.
- Low-latency internet recommended if remote updates or server-side processing is added in the future.

4. **Peripheral Devices**

- Webcam capable of at least 720p resolution for accurate facial detection.
- Microphone optional if adding audio guidance or interactive features.

Software Requirements

1. **Face-api.js** – A JavaScript library built on TensorFlow.js for performing real-time face detection, age estimation, and gender classification directly in the browser.

2. **TensorFlow.js** – Provides support for running deep learning models on client-side web applications, enabling real-time predictions without relying on external servers.
3. **Node.js** – Used to run the development server, manage packages, and handle any backend operations required during development.
4. **OpenCV.js (optional)** – Can be used for preprocessing webcam images, enhancing quality, or performing additional image transformations for improved detection accuracy.
5. **Visual Studio Code (VS Code)** – Recommended integrated development environment (IDE) for coding, debugging, and testing JavaScript, HTML, and CSS code.
6. **Git** – Version control system to manage code revisions and facilitate collaboration during development.
7. **Modern Web Browsers** – Google Chrome, Mozilla Firefox, or Microsoft Edge, required to run the web application and support WebGL and JavaScript-based AI execution.
8. **Audacity (optional)** – For recording and editing audio guidance if interactive audio feedback is added to the system.
9. **Figma (optional)** – For designing and prototyping the web interface, allowing for a user-friendly and visually intuitive UI.
10. **npm / Yarn** – Package managers for installing dependencies and managing JavaScript libraries used in the project.

Table 2 Technologies Used

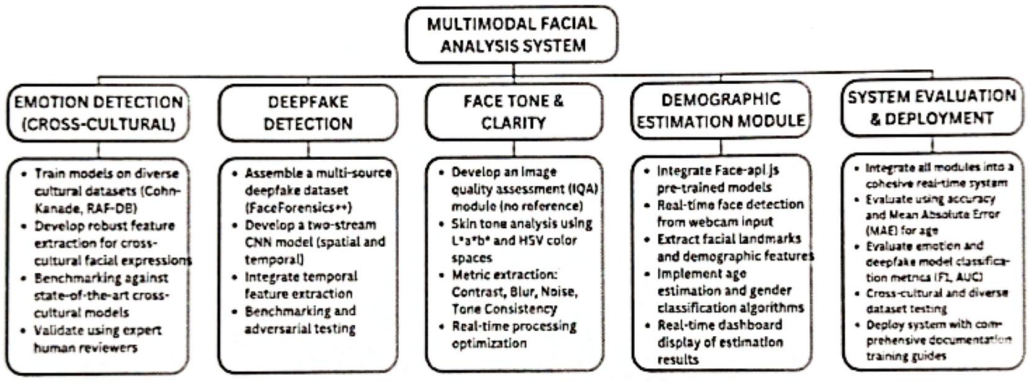
Software / Tool	Purpose / Use
Face-api.js	Run pre-trained deep learning models in the browser for face detection, age prediction, and gender classification.
TensorFlow.js	Framework for running deep learning models directly in the web browser.
Node.js	Backend services, package management, and running development server.

Software / Tool	Purpose / Use
OpenCV.js	Optional library for image preprocessing and enhancements.
Visual Studio Code	Development environment for writing, testing, and debugging JavaScript and web code.
Git	Version control and code management for collaboration and backup.
Google Chrome / Mozilla Firefox / Microsoft Edge	Modern browsers for running the web application with WebGL and JavaScript support.
Audacity	Optional, for recording/editing audio if audio guidance is implemented.
Figma	Optional, for designing and prototyping the user interface.

Figure 2 Gantt Chart

Phase	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Literature Review & Benchmarking								
AI Model Development								
Image Preprocessing & Testing								
UI/UX Design								
App Development								
Face-API testing and Lightweight model testing								
Final Validation and PWA activation								

Figure 2 Work Breakdown Structure (WBS)



Description of Personal and Facilities

Supervisor

Our supervisor, Dr Junius Anjana, and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will guide software development and technology integration.

Facilities

The project is carried out with support from the Information Technology Department of SLIIT, where access is provided to computer laboratories, academic resources, and development tools required for building and testing the application. The project utilizes available computing facilities and internet access to develop and evaluate the web-based facial analysis system. Guidance from lecturers and access to technical documentation related to artificial intelligence, computer vision, and web development support the implementation process. This environment ensures that the system is developed following standard software engineering practices while enabling effective experimentation, testing, and evaluation of the age and gender detection modules.

Device Access

The application will be tested on commonly used laptops and desktop computers equipped with webcams, focusing on devices with different performance levels to ensure the system works smoothly across a wide range of hardware. Testing will include different users interacting with the web interface to verify that face detection, age estimation, and gender classification functions operate correctly in real-time.
The system will also be evaluated under different environmental conditions such as varying lighting levels, camera resolutions, and background settings to ensure reliable performance. In addition, the web-based implementation allows the application to run directly in modern browsers without requiring complex installations, making it easily accessible to users. The system will process facial data locally within the browser,

helping to maintain user privacy while providing real-time predictions in a simple and efficient manner.

Budget and Budget Justification

This project requires a modest but strategically allocated budget to support the development, testing, and evaluation of a bilingual AI-assisted mobile application for infant cranial and posture monitoring. The budget prioritizes essential tools, infrastructure, and user engagement to ensure both technical feasibility and real-world usability.

Table 3 Budget Table

Item	Estimated Cost (LKR)	Purpose
Software Tools (Face-api.js, TensorFlow.js, VS Code, Git)	0	Open-source tools used for implementing face detection, age estimation, and gender classification.
Development Environment (Laptop / Desktop with Webcam)	0	Personal or university-provided device used for coding, testing, and running the web application.
Internet Access	2,000	Required for downloading libraries, models, and accessing documentation during development.
Testing Environment	0	Testing performed using available webcams and browsers on existing computers.
Documentation and Report Preparation	1,000	Printing, formatting, and preparing the final research report and supporting documents.
Miscellaneous / Contingency	2,000	Small buffer for unexpected development or testing expenses.

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- **Rothe, R., Timofte, R., & Van Gool, L. (2015).**
DEX: Deep EXpectation of Apparent Age from a Single Image.
IEEE International Conference on Computer Vision Workshops (ICCVW).
— Proposes a deep learning approach to estimate apparent age from facial images.
- **Guo, Y., Zhang, L., Hu, Y., He, X., & Gao, J. (2016).**
MS-Celeb-1M: A Dataset and Benchmark for Large-Scale Face Recognition.
European Conference on Computer Vision (ECCV).
— Provides a large-scale dataset often used to pre-train models for age and gender estimation.
- **Ricanek, K., & Tesafaye, T. (2006).**
MORPH: A Longitudinal Image Database of Normal Adult Age-Progression.
— Widely used dataset for age-related research and gender classification.
- **Face-api.js (2020).**
Face-api.js: JavaScript library for face detection and demographic prediction in browsers.
GitHub repository: https://github.com/justadudewhohacks/face-api.js
— Provides pre-trained models for age and gender estimation that can run directly in web browsers.
- **Smilkov, D., et al. (2019).**
TensorFlow.js: Machine Learning for the Web and Beyond.
arXiv preprint.
— Framework for running deep learning models like CNNs in the browser, enabling age and gender detection on client-side.
- **Levi, G., & Hassner, T. (2015).**
Age and Gender Classification Using Deep Learning.
CVPR Workshops.
— Focuses on model architectures and evaluation metrics specifically for demographic prediction.